

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.7: Surface Area and Volume of Solids — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.7: Surface Area and Volume of Solids
Driving Question	DRIVING QUESTION: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Surface Area of Prisms: Same Metal Sheet, Different Grain Capacity?	Learners examined a set of open metal containers of different prism shapes (rectangular, triangular, and trapezoidal cross-sections) made from the same area of metal sheet. By handling the containers and pouring maize between them, learners observed that containers with identical surface areas held noticeably different volumes of grain. This created productive confusion that anchored the unit's driving question. Teachers should cold-call at least 3 learners to articulate WHY they think this happens before any instruction begins — this surfaces prior knowledge and misconceptions that will be revisited in Lesson 5.	Learners established that surface area is the total area of all outer faces of a three-dimensional solid. For any prism, the surface area is calculated by summing the areas of every individual face: $SA = 2(\text{area of base}) + (\text{perimeter of base} \times \text{height})$. Learners practised identifying nets of rectangular and triangular prisms and mapping each face of the net to its corresponding formula component. Key vocabulary formalised: face, edge, vertex, net, lateral face, base.	The same total area of metal sheet (i.e., the same surface area) can be folded or cut into prisms of very different shapes, and the enclosed volume depends on the shape — not just the amount of material. A Kenyan jua kali artisan making grain storage boxes from galvanised iron sheets therefore cannot assume that using the same sheet size guarantees the same storage capacity. Teachers should explicitly connect this back to the anchoring phenomenon: 'We used the same metal sheet — why did the maize level differ?' Learners should record this question on the Driving Question Board as an open thread to be resolved progressively across the unit.
Lesson 2 Surface Area of Pyramids: Designing Pyramid Roofs for Kenyan Structures	Learners observed photographs and scale drawings of pyramid-roofed structures common in Kenyan rural construction (granaries, bandas, and traditional homesteads in the Rift Valley and Western Kenya). They then unfolded a paper net of a square-based pyramid and physically measured the base square and each triangular face. By recording measurements	Learners derived the surface area formula for a square-based pyramid: $SA = b^2 + 2bl$, where b is the base side length and l is the slant height. They learned to distinguish clearly between perpendicular height (h) and slant height (l), using the Pythagorean theorem to convert between them when necessary: $l = \sqrt{h^2 + (b/2)^2}$. The formula was extended to rectangular-based	A Kenyan artisan designing a pyramid-shaped granary roof must use the slant height — not the vertical height — when calculating the amount of iron sheet or thatch required to cover the roof. Using the vertical height would significantly underestimate the material needed, leading to wastage of money or an incomplete roof. Teachers should pose: 'If a Kericho tea

	and comparing peer results, learners noticed that all four triangular faces are congruent and that the height of each triangle is the slant height — NOT the perpendicular height of the pyramid. Teachers should prompt: 'What is different about the height we measure on the triangle versus the height of the full pyramid?' Allow 30 seconds wait time before taking responses.	pyramids by treating each pair of triangular faces separately.	farmer wants to roof a 3 m × 3 m square granary with a pyramid roof of vertical height 2 m, how many square metres of iron sheet are needed?' Learners solve individually, then compare answers with a partner before a whole-class check.
Lesson 3 Surface Area of Cones and Frustums: Unrolling the Cone to Find the Metal Sheet	Learners unrolled a paper cone (representing a conical grain funnel used at posho mills in Nairobi and Kisumu markets) into its flat net. They observed that the lateral surface unfolds into a sector of a circle — not a rectangle or triangle. Learners measured the radius of the sector (equal to the slant height l of the cone) and the arc length of the sector (equal to the base circumference $2\pi r$ of the cone). Teachers should ask learners to predict the formula for the area of the sector BEFORE revealing it, recording predictions on the board. Then guide the derivation collaboratively.	Learners derived that the lateral (curved) surface area of a cone equals πrl , where r is the base radius and l is the slant height. The total surface area of a closed cone is $TSA = \pi rl + \pi r^2 = \pi r(l + r)$. For a frustum (a cone with its tip removed — the shape of many Kenyan metal buckets and storage silos), the curved surface area is $\pi(R + r)l$, where R and r are the radii of the two circular ends and l is the slant height of the frustum. The total surface area of a frustum adds the two circular ends: $TSA = \pi(R + r)l + \pi R^2 + \pi r^2$.	Metal buckets and conical funnels used at grain markets (posho mills) across Kenya are frustum-shaped. A jua kali artisan in Gikomba market who fabricates these from flat metal sheet must calculate the exact curved surface area to cut the correct sector shape before rolling it. An error in using vertical height instead of slant height wastes expensive sheet metal. Teachers should highlight the real-world stakes and cold-call learners: 'Where else in Kenya do you see frustum shapes?' (Expected answers: water tanks, flower-pot shapes, silos, cooking pots.)
Lesson 4 Surface Area of Spheres and Composite Solids	Learners wrapped a spherical orange (or ball) completely with a strip of paper and then unwrapped it to measure the covered area. They then compared this to the area of four circles drawn using the same orange as a template (radius = radius of the orange). Learners observed that the covered paper area closely matched the area of four circles — visually confirming that $SA = 4\pi r^2$. For composite solids, learners assembled a model of a cylinder topped with a hemisphere (representing a common Kenyan water tank design) and identified which surfaces are external and which are hidden/internal at the join.	The surface area of a sphere is $SA = 4\pi r^2$. The surface area of a hemisphere (half-sphere) is $2\pi r^2$ (curved surface) + πr^2 (flat circular base) = $3\pi r^2$. For composite solids, learners learned to: (1) identify and sum only the EXPOSED faces of each component solid, (2) subtract any faces hidden at joints between solids, and (3) never simply add the full surface areas of each component. This systematic decomposition strategy is the key skill for composite solid surface area problems.	Many Kenyan water storage tanks are composite solids: a cylindrical body topped with a hemispherical cap. A fabricator cannot simply add the full surface area of the cylinder and the full surface area of the hemisphere — the circular face at the joint is internal and must be subtracted from both components. Teachers should emphasise this as a common error source and provide a worked example using a tank common in rural Rift Valley homesteads. Ask learners: 'Which faces disappear when we join the cylinder and hemisphere?' Cold-call three learners and annotate a diagram on the board.
Lesson 5 Volume of Prisms and Pyramids: How Much Grain Can the Box Really Hold?	Learners revisited the anchoring phenomenon from Lesson 1: the same-surface-area prism containers were now filled with dry maize and the volumes compared by counting cupfuls. Learners	Volume of any prism: $V = B \times h$, where B is the area of the base and h is the perpendicular height. Volume of a pyramid: $V = (1/3) \times B \times h$. Learners derived the pyramid formula experimentally by filling a	A Kenyan farmer storing maize in a rectangular granary (prism shape) holds three times more grain than a neighbour with a pyramid-shaped granary of the same base area and height — even if both

	recorded the base dimensions and heights of each prism and calculated the base area. They observed that the container with the largest base area $\times$ height held the most grain, even though all containers were made from identical amounts of metal sheet. This observation directly resolves the anchoring question for prisms and sets up the volume-optimisation theme.	pyramid-shaped container with sand and pouring it into a prism of identical base and height — exactly three fillings were required. This 1:3 ratio between pyramid and prism volumes (same base and height) must be memorised and its derivation understood. Composite prism and pyramid volumes are found by summing component volumes.	structures use the same amount of timber or iron sheet. This directly answers the driving question for prisms and pyramids. Teachers should ask: 'Which granary shape gives more storage per unit of construction material — prism or pyramid?' and discuss the trade-offs (cost of materials vs. storage capacity). Connect to real farming decisions made in Uasin Gishu or Trans-Nzoia Counties.
Lesson 6 Volume of Cylinders, Cones, and Spheres	Learners filled a cylindrical tin (e.g., a standard Kimbo or Afia tin) with water and poured it into a cone of identical base radius and height, observing that the cylinder filled the cone exactly three times — mirroring the prism-pyramid relationship from Lesson 5. They then observed a spherical water container (or ball) and estimated its volume by submerging it in a rectangular tank and measuring the water displaced (Archimedes' principle), comparing the displaced volume to the formula prediction.	Volume of a cylinder: $V = \pi r^2 h$. Volume of a cone: $V = (1/3)\pi r^2 h$ — exactly one-third of the cylinder with the same base and height. Volume of a sphere: $V = (4/3)\pi r^3$. Volume of a hemisphere: $V = (2/3)\pi r^3$. Learners should commit all four formulae to memory and understand the structural relationship between cylinder/cone and prism/pyramid (the 1:3 ratio is consistent). Practice problems should include mixed units (converting cm^3 to litres: 1 litre = 1000 cm^3) as this is essential for real-world grain and water storage contexts.	Cylindrical metal containers (drums) are the most common grain and water storage vessels in Kenya because they maximise volume relative to material used and are easy to fabricate by rolling flat sheet metal. A standard 200-litre oil drum holds far more than a cone or pyramid made from the same metal sheet. Teachers should pose: 'A posho mill owner in Eldoret wants to store 500 litres of maize flour. How many cylindrical drums of radius 30 cm and height 90 cm does she need?' This applied problem ties the formula directly to realistic Kenyan commercial contexts. Learners should convert cm^3 to litres in their final answer.
Lesson 7 Volume and Surface Area of Composite Solids & Real-Life Applications	Learners examined diagrams and photographs of composite solid structures common in Kenya: a cylindrical water tank with a conical base (used in Nairobi estates), a silo with a cylindrical body and hemispherical top, and a grain store with a rectangular prism body and pyramid roof. For each, learners identified the component solids, marked the boundaries between components, and listed which faces are exposed (contributing to surface area) and which are hidden (excluded from surface area but not from volume).	Learners learned that: (1) a composite solid is decomposed into named component solids; (2) total VOLUME of a composite solid is always the SUM of component volumes (no faces are 'hidden' from volume); (3) total SURFACE AREA requires careful subtraction of any faces concealed at internal joints — the rule is to add all component surface areas then subtract twice the area of each shared internal face; (4) real-life problems often require unit conversions (cm^3 to litres, m^2 to cm^2) and selecting appropriate levels of precision ($\pi \approx 3.142$ or use calculator).	A Kenyan water-tank fabricator designing a tank with a cylindrical body ($r = 0.5 \text{ m}$, $h = 1.2 \text{ m}$) and a hemispherical top ($r = 0.5 \text{ m}$) must calculate both the total volume (for rated capacity in litres) and the total external surface area (for the amount of metal sheet needed). Failing to subtract the shared circular face at the cylinder-hemisphere joint causes a costly overestimate of sheet metal. Teachers should work through one full composite-solid problem on the board — narrating every decomposition, addition, and subtraction step aloud — before releasing learners to solve a parallel problem in pairs. Cold-call pairs to present steps, not just final answers.
Lesson 8 Project Presentations,	Learners presented their project designs — annotated drawings or physical models of a	Learners synthesised the full unit's content into a final annotated model consolidating	The driving question is fully resolved: the same amount of metal sheet (same surface

Final Model Building & Unit Consolidation	storage or construction solution for a Kenyan artisan (e.g., a grain silo, water tank, or roofed granary). Each presentation included: the chosen solid shape(s), all measured or assumed dimensions, surface area calculations (materials needed), and volume calculations (storage capacity). Peers observed each presentation and provided structured feedback using a provided rubric: accuracy of formulae used, correctness of unit conversions, clarity of real-world justification, and quality of the annotated diagram.	all formulae: prisms ($SA = 2B + Ph$; $V = Bh$), pyramids ($SA = B + \frac{1}{2}Pl$; $V = \frac{1}{3}Bh$), cylinders ($TSA = 2\pi r^2 + 2\pi rh$; $V = \pi r^2h$), cones ($TSA = \pi r(l + r)$; $V = \frac{1}{3}\pi r^2h$), spheres ($SA = 4\pi r^2$; $V = \frac{4}{3}\pi r^3$), frustums (curved $SA = \pi(R+r)l$), and composite solids (volume = sum of parts; surface area = sum of parts minus hidden faces). The unifying insight of the unit — that surface area governs material cost while volume governs storage capacity, and these two quantities do not scale together — was stated explicitly and recorded.	area) makes containers of very different volumes because volume depends on the shape and dimensions in a non-linear way (involving squared and cubed terms), while surface area grows more slowly. A Kenyan artisan who understands this mathematics can choose the optimal shape: a cylinder maximises volume for a given surface area among common fabrication-friendly shapes, which is why cylindrical drums dominate grain and water storage across Kenya. Teachers should close by asking learners to reflect: 'Has your answer to the driving question changed since Lesson 1?' and have three volunteers share their intellectual journey. Record the final resolved answer on the Driving Question Board alongside the initial predictions from Lesson 1.
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